# 1 Introduction

Sometimes a given basis is not the most convenient for a particular problem, and it would be better to change to another basis. Consider, for example, the motion of an object down an inclined plane angled $\theta = 30^{\text{deg}}$ above the ground.
Since the basis vectors of the standard basis of $\mathbf{R}^2$ are parallel and perpendicular to the ground, there is motion in both of these directions, so Newton's second law gives rise to a coupled system of differential equations. It is much more convenient to choose a rotated coordinate system, which corresponds to choosing the basis

$$v_{\parallel} = (\cos(30^{\text{deg}}), \sin(30^{\text{deg}})) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$
$$v_{\perp} = (\sin(30^{\text{deg}}), \cos(30^{\text{deg}})) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

where $v_{\parallel}$ points down the plane and $v_{\perp}$ is perpendicular to the plane. It is much easier to solve the equations of motion in this basis since the motion is only along the $v_{\parallel}$ direction.

# 2 Change of Basis - Coordinate Vectors

Let $V$ be a finite-dimensional vector space and let $B = (v_1, ..., v_n)$ be an ordered basis of $V$. Then for any $v \in V$, there exist unique scalars $\alpha_1, ..., \alpha_n$, such that

$$v = \alpha_1 v_1 + ... + \alpha_n v_n. \tag{1}$$

Let $B' = (w_1, ..., w_n)$ be another ordered basis of $V$. There is another set of unique scalars $\beta_1, ..., \beta_n$ such that

$$v = \beta_1 w_1 + ... + \beta_n w_n \tag{2}$$

We would like to know how the coordinate vector $[v]_{B'} = (\beta_1, ..., \beta_n)$ is obtained from $[v]_B = (\alpha_1, ..., \alpha_n)$ by changing basis from $B$ to $B'$.
We can write each vector $v \in V$ as a linear combination of the vectors in $B'$. In particular, this is true for each vector in $B$. That is, for each $j = 1, ..., n$, there exist unique scalars $C_{ij}$ such that

$$v_j = C_{1j} w_1 + ... + C_{nj} w_j. \tag{3}$$

Substituting (2.3) into (2.1) for each $v_j$ gives

$$v = \sum_{j=1}^{n} \alpha_j \sum_{i=1}^{n} C_{ij} w_i = \sum_{i=1}^{n} \left( \sum_{j=1}^{n} C_{ij} \alpha ij \right) w_i,$$

and using the fact that the $w_i$ form a basis and using 2.2, we have:

$$B_i = \sum_{j=1}^{n} C_{ij} \alpha_j.$$

That is, to obtain the new coordinate vector $[v]_{B'}$, we simply multiply the old coordinate vector $[v]_B$ by the matrix $C$, whose $j$th column is $[v_j]_{B'}$ (the coordinate vector of the $j$th old basis vector with respect to the new basis $B'$). The matrix $C$ is called the transition matrix to change of basis matrix from $B$ to $B'$.
By interchanging the roles of $B$ and $B'$, it follows immediately that $C$ is invertible, with $C^{-1}$ being the transition matrix from $B'$ to $B$.



**Example 2.1**

Let $B = (v_1, v_2) = ((1, 0), (0, 1))$ be the standard basis of $\mathbf{R}^2$ and $B' = (w_1, w_2) = ((\sqrt{3}/2, 1/2), (-1/2, \sqrt{3}/2))$ be the rotated basis from the example discussed in the first section. To find the matrix $C$, we expand

each basis vector $v_j$ of $B$ in the basis $B'$.

$$\begin{aligned} v_1 &= C_{11}w_1 + C_{21}w_2 \\ (1,0) &= C_{11}\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) + C_{21}\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \\ (1,0) &= \left(\frac{\sqrt{3}}{2}C_{11} - \frac{1}{2}C_{21}, \frac{1}{2}C_{11} + \frac{\sqrt{3}}{2}C_{21}\right) \end{aligned}$$

which implies the system of linear equations

$$\begin{aligned} \frac{\sqrt{3}}{2}C_{11} - \frac{1}{2}C_{21} &= 1 \\ \frac{1}{2}C_{11} + \frac{\sqrt{3}}{2}C_{21} &= 0 \end{aligned}$$

and the second column is given by solving:

$$\begin{aligned} v_2 &= C_{21}w_1 + C_{22}w_2 \\ (0,1) &= C_{21}\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) + C_{22}\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \\ (0,1) &= \left(\frac{\sqrt{3}}{2}C_{21} - \frac{1}{2}C_{22}, \frac{1}{2}C_{21} + \frac{\sqrt{3}}{2}C_{22}\right), \end{aligned}$$

implying the system:

$$\begin{aligned} \frac{\sqrt{3}}{2}C_{21} - \frac{1}{2}C_{22} &= 0 \\ \frac{1}{2}C_{21} + \frac{\sqrt{3}}{2}C_{22} &= 1. \end{aligned}$$

Notice that these two systems have exactly the same coefficient matrix. We can therefore solve both systems simultaneously by row-reduction by forming an augmented matrix with two additional columns $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$:

$$\left[\begin{array}{cc|cc} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 1 \end{array}\right]$$

Row reducing gives:

$$\begin{pmatrix} 1 & 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

and therefore

$$C = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

**Remark 2.2**

1. More generally, to find the transition matrix $C$ from basis $B$ to $B'$, we row-reduce the augmented matrix $[[w_i] \mid [v_i]]$, whose first $n$ columns are the coordinate vectors $[w_i]$ of the vectors in $B'$ and whose last $n$ columns are the coordinate vectors $[v_i]$ of the vectors in $B$.
2. Row-reducing the augmented matrix in (2.5), as we learned last time, is precisely the technique for computing the inverse, so

$$C^{-1} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

**Example 2.3**

Let $B, B'$ be ordered bases for $F^3$ where $v_1 = (2,1,1)$, $v_2 = (2,-1,1)$, $v_3 = (1,2,1)$ and $w_1 = (3,1,-5)$, $w_2 = (1,1,-3)$, $2_3 = (-1,0,2)$. To compute the transition matrix $C$ from $B$ to $B'$, we row-reduce the augmented matrix

$$[[w_i] \mid [v_j]] = \left[\begin{array}{ccc|ccc} 3 & 1 & -1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & -1 & 2 \\ -5 & -3 & 2 & 1 & 1 & 1 \end{array}\right] \to \begin{pmatrix} 1 & 0 & 0 & 3 & 2 & \frac{5}{2} \\ 0 & 1 & 0 & -2 & -3 & -\frac{1}{2} \\ 0 & 0 & 1 & 5 & 1 & 6 \end{pmatrix}$$

and therefore $C = \begin{bmatrix} 3 & 2 & \frac{5}{2} \\ -2 & -3 & -\frac{1}{2} \\ 5 & 1 & 7 \end{bmatrix}$.

# 3 Change of Basis - Matrix of a Linear Operator

Let $T = \mathcal{L}(V)$ and $B, B'$ two ordered bases of a finite-dimensional vector space $V$. Let $w = Tv$. In the coordinates defined by $B$, this becomes

$$[w]_B = [T]_b[v_b] \tag{4}$$

We found in the previous section that $[w]_{B'} = C[w_b]$ and $[v]_{B'} = C[v]_B$, so if we instead use the coordinates defined by $B'$ we have

$$\begin{aligned} [w]_{B'} &= [T]_{B'}[v]_{B'} \\ C[w]_B &= [T]_{B'}C[v]_{B'}. \end{aligned}$$

Multiplying both sides by $C^{-1}$ then gives:

$$[w]_B = C^{-1}[T]_{B'}C[v]_B. \tag{5}$$

Comparing (3.2) and (3.1), we see that $[T]_B$ and $[T]_{B'}$ are related by

$$C^{-1}[T]_{B'}C = [T]_B,$$

or, multiplying on the left by $C$ and the right by $C^{-1}$,

$$[T]_{B'} = C[T]_BC^{-1}. \tag{6}$$

**Example 3.1**

Consider the linear map $T : \mathbf{R}^2 \to \mathbf{R}^2$ defined by $T(x,y) = (2x-y, -3x+2y)$. With respect to the standard ordered basis $B$ of $\mathbf{R}^2$, the matrix of $T$ is given by

$$[T]_B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

If we change basis to the ordered basis $B' = \{\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, we found in 2.4 that

$$C = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \text{ and } C^{-1} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix},$$

and therefore the matrix of $T$ with respect to the new ordered basis $B'$ is

$$\begin{aligned} [T]_{B'} &= C[T]_B C^{-1} \\ &= \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}2}{} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \\ &= \begin{bmatrix} 2-\sqrt{3} & 0 \\ -2 & 2+\sqrt{3} \end{bmatrix}. \end{aligned}$$

# 4 Exercises



**Problem 1**

Consider the ordered bases $B = (v_1, v_2)$ and $B$" $= (w_1, w_2)$ of $F^2$ where

$$v_1 = (2, 2), v_2 = (4, -1)$$
$$w_1 = (1, 3), w_2 = (-1, -1).$$

(a) Find the transition matrix from $B$ to $B'$.

(b) Find the transition matrix from $B'$ to $B$.

(c) Given $u = (3, -5)$ compute the coordinate vectors $[u]_B$ and $[u]_{B'}$.

(d) Given the linear operator $T(x, y) = (x + 2y, -4x - 2y)$, calculate $[T]_B$ and $[T]_{B'}$.



**Solution**

(a) We need to row reduce:

$$\left[\begin{array}{cc|cc} 1 & -1 & 2 & 4 \\ 3 & -1 & 2 & -1 \end{array}\right]$$

which gives:

$$\left[\begin{array}{cc|c} 1 & 0 & 0 & -\frac{5}{2} \\ 0 & 1 & -2 & -\frac{13}{2} \end{array}\right]$$

So, the transition matrix is:

$$\begin{bmatrix} 0 & -\frac{5}{2} \\ -2 & -\frac{13}{2} \end{bmatrix}$$

(b) The other transition matrix is the inverse of this, which we get by row-reducing:

$$\left[\begin{array}{cc|cc} 0 & -\frac{5}{2} & 1 & 0 \\ -2 & -\frac{13}{2} & 0 & 1 \end{array}\right]$$

to get

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{13}{10} & -\frac{1}{2} \\ 0 & 1 & -\frac{2}{5} & 0 \end{array}\right]$$

so our transition matrix is:

$$\begin{bmatrix} \frac{13}{10} & -\frac{1}{2} \\ -\frac{2}{5} & 0 \end{bmatrix}$$

(c) The transition matrix from the standard basis to $B$ is given by the RHS of row-reducing:

$$\left[\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ 2 & -1 & 0 & 1 \end{array}\right]$$

which gives:

$$\begin{bmatrix} \frac{1}{10} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{bmatrix}$$

and so $(3, 5)$ goes to:

$$\begin{bmatrix} \frac{23}{10} \\ -\frac{2}{5} \end{bmatrix}$$

and applying the transition to $B'$ gives:

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

(d) $T_b$ in the standard basis looks like:

$$\begin{bmatrix} 1 & 2 \\ -4 & -2 \end{bmatrix}$$

The map from $B$ to the standard basis is of course:

$$\begin{bmatrix} 2 & 4 \\ 2 & -1 \end{bmatrix}$$

so $T_B$ goes to:

$$\begin{bmatrix} \frac{1}{10} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{21}{5} & -\frac{27}{5} \\ \frac{18}{5} & \frac{16}{5} \end{bmatrix}$$

and then conjugating to go to $B'$ gives:

$$\begin{bmatrix} 0 & -\frac{5}{2} \\ -2 & -\frac{13}{2} \end{bmatrix} \begin{bmatrix} -\frac{21}{5} & -\frac{27}{5} \\ \frac{18}{5} & \frac{16}{5} \end{bmatrix} \begin{bmatrix} \frac{13}{10} & -\frac{1}{2} \\ -\frac{2}{5} & 0 \end{bmatrix} = \begin{bmatrix} -\frac{17}{2} & \frac{9}{2} \\ -\frac{31}{2} & \frac{15}{2} \end{bmatrix}$$